



**Department of Information and Communication Technology
Faculty of Science and Technology
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Project Proposal

BUP Research Bridge

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Abstract

BUP Research Bridge is a centralized web-based research platform proposed for the Computer Science and Engineering (CSE) and Information and Communication Technology (ICT) departments of Bangladesh University of Professionals (BUP). The platform is designed to address the difficulties students face in finding suitable research supervisors, thesis opportunities, faculty research interests, previous thesis works, and experienced alumni. Currently, research-related information is scattered across multiple sources, making the discovery process inefficient and time-consuming.

The proposed system will provide a unified platform where students can explore faculty profiles, research interests, publications, available thesis opportunities, and previous research projects. Faculty members will be able to manage their research profiles, publish research opportunities, and indicate their availability for supervision. Alumni will also be able to share their research experiences, publications, and areas of expertise, creating stronger connections between current students and graduates.

The project will be developed using an Agile Scrum methodology and implemented using modern web technologies. The expected outcome is an organized research ecosystem that improves accessibility to research resources, strengthens collaboration among students, faculty members, and alumni, and supports research growth within the BUP community.

Keywords: Research Portal, Thesis Management, Faculty Mentorship, Research Collaboration, Alumni Network

1 Introduction

Research plays a significant role in academic and professional development, particularly in higher education institutions where students engage in thesis work, publications, and collaborative projects. At Bangladesh University of Professionals (BUP), students from the Computer Science and Engineering (CSE) and Information and Communication Technology (ICT) departments often face difficulties in identifying suitable research supervisors, understanding faculty research interests, discovering available thesis opportunities, and accessing information about previous research works. Similarly, faculty members have limited centralized channels to showcase their research expertise, announce research opportunities, and connect with interested students.

Currently, research-related information is distributed across departmental websites, personal faculty profiles, social media groups, academic publications, and informal communication channels. This fragmented structure makes it difficult for students to efficiently explore

research opportunities and establish connections with potential supervisors. In addition, alumni who have valuable research experience and academic achievements often remain disconnected from current students, limiting opportunities for mentorship and collaboration.

To address these challenges, the proposed system, BUP Research Bridge, aims to provide a centralized web-based platform that connects students, faculty members, and alumni within the BUP research community. The platform will serve as a research hub where users can access faculty research profiles, publications, thesis opportunities, previous thesis projects, and alumni achievements. Through this system, students will be able to make informed decisions regarding research topics and supervisors, while faculty members can efficiently communicate their research interests and available opportunities.

The proposed platform will strengthen research collaboration, improve information accessibility, and create a structured environment that supports academic research activities within BUP.

2 Problem Statement

Students in the CSE and ICT departments of Bangladesh University of Professionals often face significant challenges when searching for research supervisors, thesis opportunities, and relevant research domains. Information regarding faculty expertise, current research activities, available thesis positions, and previous research projects is frequently scattered across multiple sources, making the process time-consuming and inefficient.

Faculty members also face difficulties in communicating their research interests and available opportunities to students in a structured manner. As a result, students may miss valuable opportunities, while faculty members may struggle to identify suitable candidates for research projects and thesis supervision.

Furthermore, alumni who have accumulated valuable research experience, academic achievements, and professional expertise remain largely disconnected from the current student community. This limits opportunities for mentorship, guidance, and collaboration between alumni and current students.

The absence of a centralized research management and collaboration platform reduces the visibility of research opportunities, creates communication gaps among stakeholders, and hinders the development of a strong research culture within the university. Therefore, a dedicated software solution is needed to centralize research-related information, improve accessibility, and facilitate collaboration among students, faculty members, and alumni.

3 Objectives

1. To design and develop a centralized research platform for the CSE and ICT departments of BUP.

2. To provide a searchable directory of faculty research profiles, interests, expertise, and publications.
3. To enable faculty members to publish and manage thesis and research opportunities.
4. To help students identify suitable research supervisors and research domains.
5. To maintain a repository of previous thesis projects and research works.
6. To provide alumni with a platform to showcase their research achievements and professional expertise.
7. To facilitate communication and collaboration among students, faculty members, and alumni.
8. To provide advanced search and filtering facilities based on research interests and expertise.
9. To improve the accessibility and visibility of research-related information within BUP.
10. To support the development of a stronger research culture within the university.
11. To evaluate the effectiveness, usability, and performance of the proposed platform.

4 Scope of the Project

The proposed system focuses on creating a centralized research platform for the Computer Science and Engineering (CSE) and Information and Communication Technology (ICT) departments of Bangladesh University of Professionals (BUP). The platform aims to improve accessibility to research-related information and facilitate collaboration among students, faculty members, and alumni.

4.1 In-Scope Features

- Faculty profile management and research interest publication.
- Thesis and research opportunity announcements.
- Research publication and project showcase.
- Previous thesis repository management.
- Alumni profile and research achievement management.
- Student access to faculty and alumni research information.

- Advanced search and filtering facilities.
- User authentication and role-based access control.
- Research collaboration and communication support.
- Administrative management of users and platform content.

4.2 Target Users

- Current BUP CSE Students
- Current BUP ICT Students
- CSE Faculty Members
- ICT Faculty Members
- CSE and ICT Alumni
- System Administrators

4.3 Supported Platforms

- Desktop Web Browsers
- Laptop Web Browsers
- Tablet Web Browsers
- Mobile Web Browsers

4.4 Out of Scope

- Full research paper submission and review management.
- Conference management functionalities.
- Integration with external journal submission systems.
- Real-time video conferencing facilities.
- Financial transaction and payment processing.
- Mobile application development during the initial phase.

StakeholdersF

5 Stakeholders

Stakeholder	Role	Benefits
Students	Search research opportunities, explore faculty profiles, review previous thesis projects, and connect with mentors.	Easy access to research information, better supervisor selection, and improved research guidance.
Faculty Members	Manage research profiles, publish thesis opportunities, supervise students, and showcase publications.	Efficient communication with students and increased visibility of research activities.
Alumni	Share research experience, publications, professional expertise, and mentorship opportunities.	Maintain academic connections and contribute to future researchers.
Administrators	Manage users, content, access control, and system maintenance.	Efficient platform administration and quality assurance.
Department Authority	Monitor research activities and support academic research initiatives.	Improved research visibility and stronger research culture.

6 Project Requirements

The proposed system shall provide a comprehensive research management and collaboration environment for students, faculty members, alumni, and administrators. The requirements are categorized into functional requirements, which describe the services and operations provided by the system, and non-functional requirements, which define quality attributes such as performance, security, reliability, usability, and maintainability.

6.1 Functional Requirements

FR-01: The system shall allow students, faculty members, alumni, and administrators to register and create accounts.

FR-02: The system shall authenticate users before granting access to protected resources.

FR-03: The system shall provide role-based access control for students, faculty members, alumni, and administrators.

FR-04: The system shall allow users to securely log in and log out.

- FR-05:** The system shall allow faculty members to create and manage research profiles.
- FR-06:** The system shall allow faculty members to update research interests and areas of expertise.
- FR-07:** The system shall allow faculty members to upload and manage publication records.
- FR-08:** The system shall allow faculty members to publish thesis and research opportunities.
- FR-09:** The system shall allow faculty members to update the status of available opportunities.
- FR-10:** The system shall allow students to browse faculty profiles.
- FR-11:** The system shall allow students to search faculty members based on research interests and expertise.
- FR-12:** The system shall allow students to view available thesis and research opportunities.
- FR-13:** The system shall allow students to browse previous thesis projects and research works.
- FR-14:** The system shall provide a searchable repository of previous thesis projects.
- FR-15:** The system shall allow alumni to create and manage professional profiles.
- FR-16:** The system shall allow alumni to publish research achievements and publications.
- FR-17:** The system shall allow alumni to share academic and professional experiences.
- FR-18:** The system shall allow students to browse alumni profiles.
- FR-19:** The system shall allow users to search research content using keywords.
- FR-20:** The system shall provide advanced filtering based on research domains, expertise, publications, and opportunities.
- FR-21:** The system shall provide profile viewing functionality for all authorized users.
- FR-22:** The system shall allow administrators to manage user accounts.
- FR-23:** The system shall allow administrators to approve, update, or remove platform content.
- FR-24:** The system shall allow administrators to manage faculty, student, and alumni records.
- FR-25:** The system shall maintain a centralized repository of research-related information.
- FR-26:** The system shall allow authorized users to update profile information.
- FR-27:** The system shall notify users about newly published research opportunities.
- FR-28:** The system shall maintain activity logs for administrative purposes.
- FR-29:** The system shall support communication between students and faculty members.
- FR-30:** The system shall support communication between students and alumni.
- FR-31:** The system shall generate administrative reports regarding platform activities.
- FR-32:** The system shall provide secure access to all research-related resources.

6.2 Non-functional Requirements

NFR-01: The system shall respond to user requests within three seconds under normal operating conditions.

NFR-02: The system shall support at least 500 concurrent users.

NFR-03: The system shall ensure 99% system availability.

NFR-04: The system shall provide secure password storage using encryption techniques.

NFR-05: The system shall enforce authentication and authorization mechanisms.

NFR-06: The system shall protect user data against unauthorized access.

NFR-07: The system shall maintain data integrity and consistency.

NFR-08: The system shall support regular database backup and recovery.

NFR-09: The system shall provide an intuitive and user-friendly interface.

NFR-10: The system shall be compatible with modern web browsers.

NFR-11: The system shall support responsive web design.

NFR-12: The system shall be scalable to accommodate future users and features.

NFR-13: The system shall maintain modular and maintainable code architecture.

NFR-14: The system shall support future integration with additional academic systems.

NFR-15: The system shall provide proper error handling and user feedback.

NFR-16: The system shall maintain comprehensive system logs.

NFR-17: The system shall provide reliable operation with minimal downtime.

NFR-18: The system shall ensure portability across different deployment environments.

NFR-19: The system shall support future enhancement and feature expansion.

7 Proposed Methodology

The development of BUP Research Bridge will follow the Agile Scrum software development methodology. Agile Scrum is selected because it supports iterative development, continuous feedback, flexibility, and effective team collaboration. The methodology enables the development team to divide the project into smaller and manageable iterations called sprints, allowing regular evaluation and improvement throughout the project lifecycle.

The project team will conduct weekly meetings to review progress, discuss challenges, and plan upcoming tasks. Project activities will be divided among team members based on expertise and workload. GitHub will be used for version control and collaboration, ensuring proper tracking of code changes and efficient team coordination.

The project development process will include requirement analysis, system design, database design, implementation, testing, deployment, and documentation. Continuous testing and review activities will be performed throughout the development cycle to ensure software quality and reliability.

7.1 Technology Stack

Category	Technology	Justification
Frontend	React.js	Provides reusable components, responsive user interfaces, and efficient state management.
Backend	Node.js with Express.js	Supports scalable server-side development and RESTful API implementation.
Database	PostgreSQL	Provides reliability, strong data integrity, and efficient handling of structured data.
API Communication	REST API	Enables standardized communication between frontend and backend systems.
Version Control	Git and GitHub	Facilitates collaboration, code management, and version tracking.
Development Tools	VS Code	Provides an efficient development environment with extensive extension support.
Deployment	Cloud Hosting Platform	Supports accessibility, scalability, and online availability of the platform.

7.2 System Architecture

Provide a high-level overview of the proposed software architecture, showing how the major components of the system communicate. Include the client (user interface), server (application logic), database, and any external services (if applicable). Explain the role of each component and briefly describe the flow of information between them. A simple architecture diagram should accompany the description. Detailed implementation or design is not required at this stage.

7.3 Preliminary System Design

The preliminary system design provides an initial conceptual representation of the proposed platform and its major functionalities. The design includes a Use Case Diagram, Context Diagram, and Basic Workflow Diagram that illustrate user interactions and information flow within the system.

The Use Case Diagram identifies the primary actors of the system, including students, faculty members, alumni, and administrators, and demonstrates their interactions with various system functionalities.

The Context Diagram presents the overall interaction between the platform and external entities, highlighting the flow of information among stakeholders.

The Workflow Diagram illustrates the operational flow of the platform, showing how users access information, manage profiles, publish research opportunities, and interact with other stakeholders.

The diagrams will be included and refined during the later stages of the project proposal.

7.3.1 Entity Relationship (ER) Diagram

The Entity Relationship (ER) Diagram illustrates the database structure of the proposed BUP Research Bridge platform. It represents the entities, attributes, and relationships among students, faculty members, alumni, research areas, thesis opportunities, publications, and mentorship requests. The diagram provides a conceptual view of the database design and serves as the foundation for database implementation.

7.3.2 Sequence Diagram

The Sequence Diagram illustrates the interaction among system components during the mentorship request process. It demonstrates how a student searches for faculty members, views research opportunities, submits mentorship requests, and receives responses from faculty members or alumni through the system.

7.3.3 Activity Diagram

The Activity Diagram represents the workflow of the mentorship request process. It describes the sequence of activities performed by students, the system, and faculty members or alumni, from searching research opportunities to receiving approval or rejection notifications.

8 Project Timeline

The project will be developed following a structured ten-week schedule consisting of requirement analysis, system design, implementation, testing, deployment, documentation, and final presentation activities. The detailed project timeline is illustrated in Figure 4.

9 Expected Outcome

The successful implementation of BUP Research Bridge is expected to provide a centralized and structured platform for research-related activities within the CSE and ICT departments of Bangladesh University of Professionals.

The platform will simplify the process of discovering research opportunities, identifying suitable supervisors, and accessing faculty research information. Students will be able to make

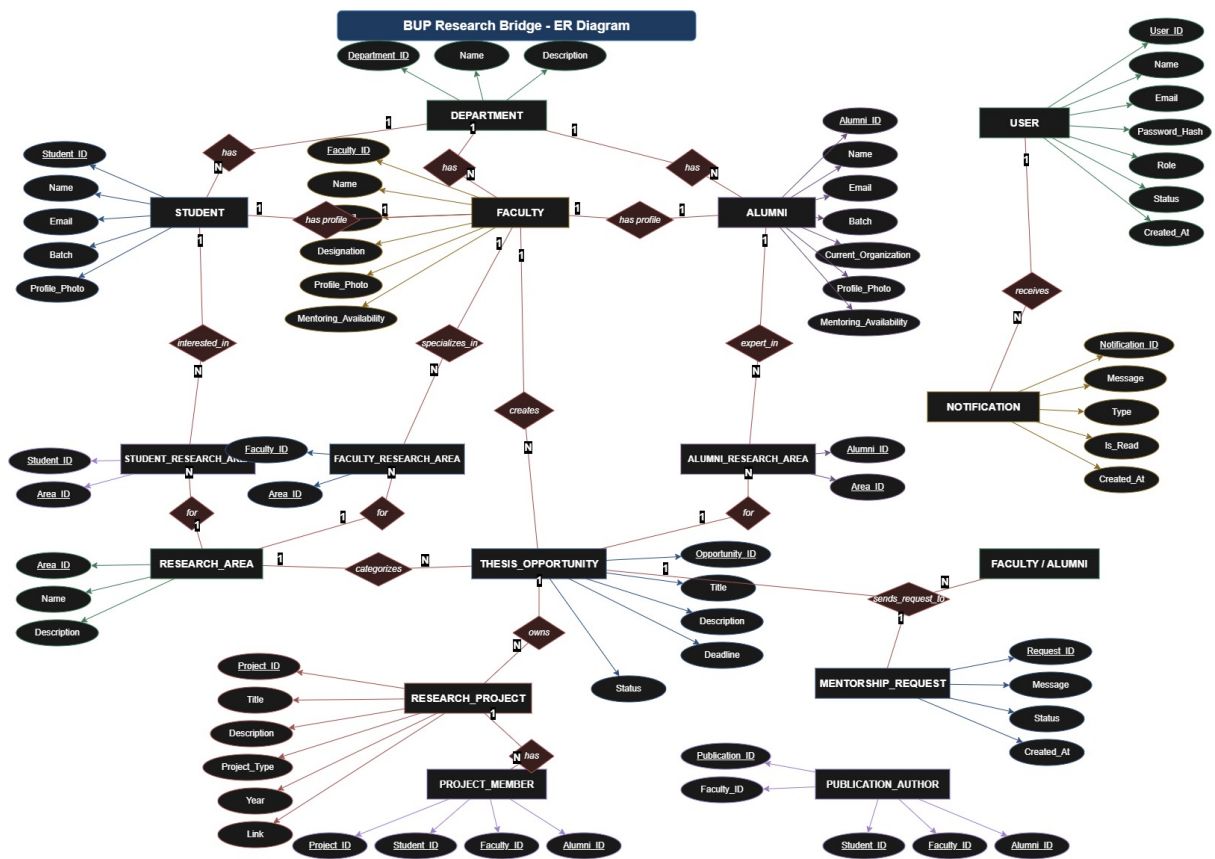
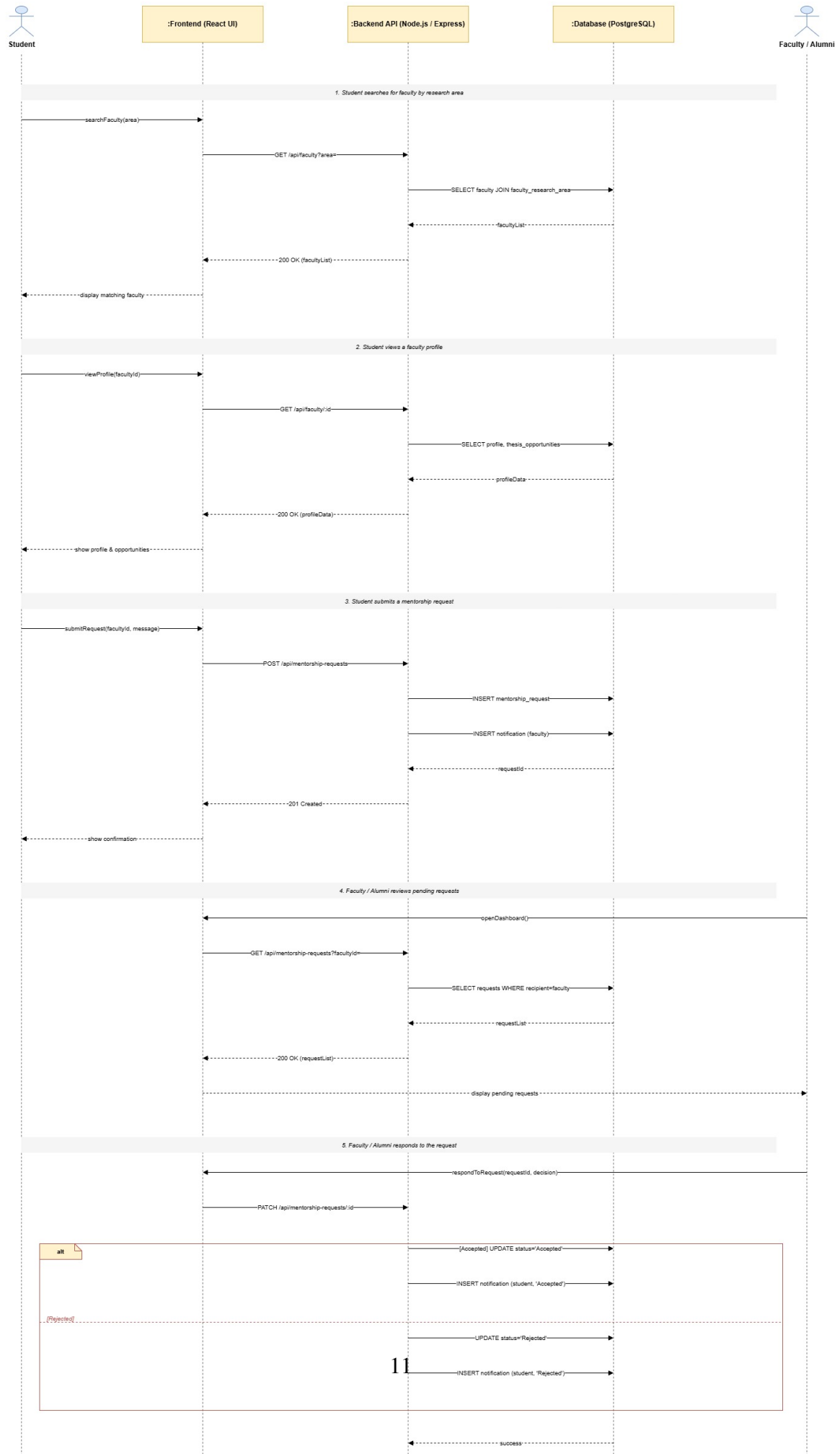


Figure 1: Entity Relationship (ER) Diagram of BUP Research Bridge

BUP Research Bridge – Sequence Diagram (Send & Respond to Mentorship Request)



BUP Research Bridge – Activity Diagram (Mentorship Request Workflow)

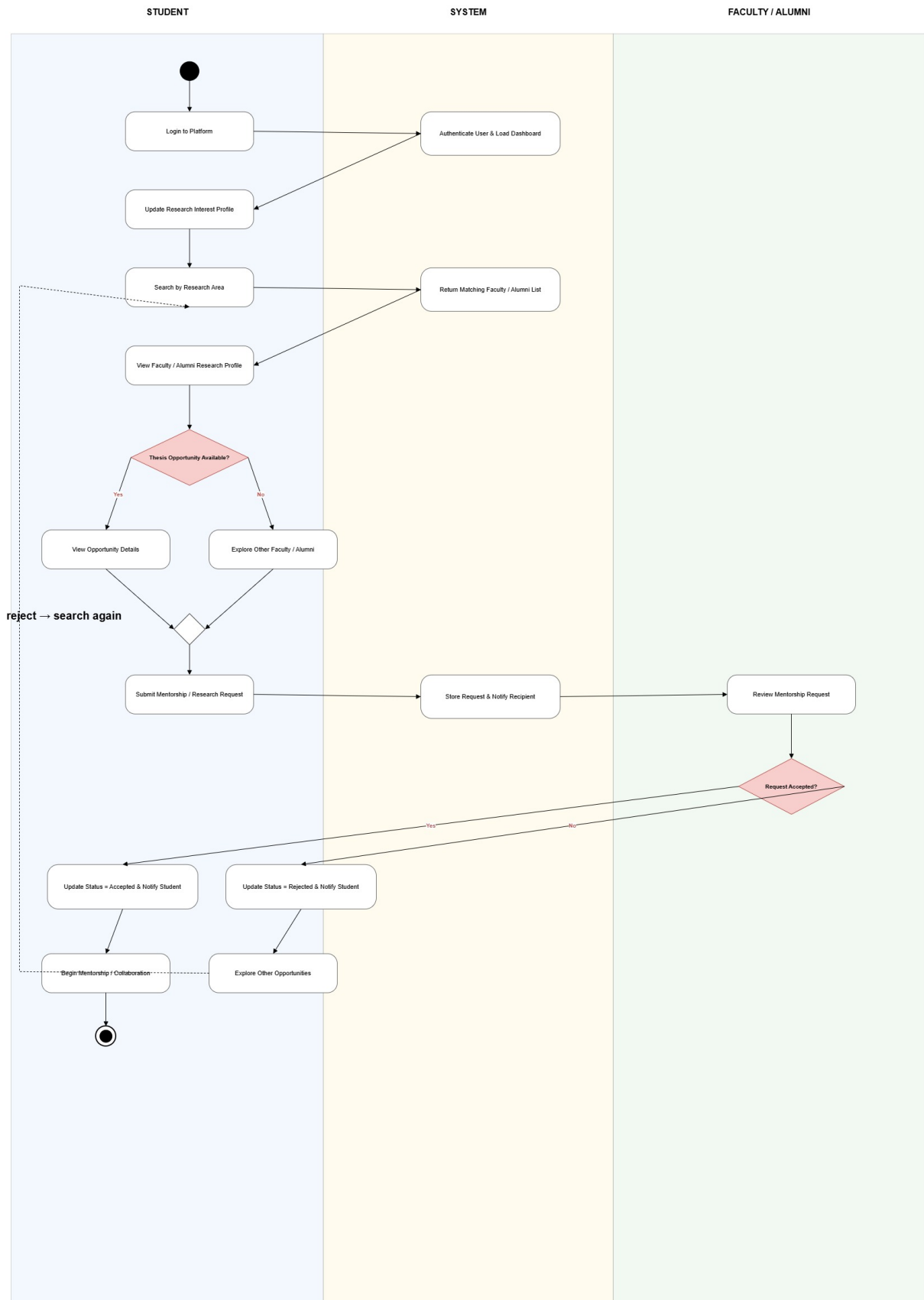


Figure 3: Activity Diagram for Mentorship Request Workflow

Gantt Chart



Figure 4: Project Gantt Chart

informed decisions regarding thesis and research work by exploring faculty expertise, previous thesis projects, and available research opportunities.

Faculty members will benefit from an organized platform where they can showcase their research interests, publications, and available thesis opportunities. This will improve visibility and facilitate effective interaction with students interested in research activities.

Alumni will be able to maintain academic connections with the university by sharing their research experiences, publications, and professional achievements. This will create opportunities for mentorship, networking, and knowledge sharing among current students and graduates.

The proposed system is expected to strengthen collaboration among students, faculty members, and alumni while promoting a stronger research culture within the university. Ultimately, the platform will contribute to improved accessibility, transparency, and efficiency in research-related information management and collaboration.

10 Literature Review

Research collaboration platforms play an important role in supporting academic research activities by providing access to research resources, publications, collaboration opportunities, and

mentorship networks. Several existing systems and technologies were reviewed to understand the requirements and design considerations for the proposed platform.

Agile Scrum is one of the most widely used software development methodologies because of its flexibility, iterative development process, and continuous feedback mechanism. Scrum enables development teams to divide projects into manageable sprints and continuously improve system functionality throughout the development lifecycle [1].

Modern web-based applications commonly utilize component-based frontend frameworks to provide responsive and interactive user experiences. React.js is a widely adopted JavaScript library that supports reusable components, efficient state management, and scalable user interface development [2].

Database management systems are essential for storing and managing structured information. PostgreSQL is a robust open-source relational database system known for reliability, data integrity, scalability, and advanced querying capabilities. These characteristics make PostgreSQL suitable for managing research-related data, user profiles, publications, and thesis information within the proposed platform [3].

Security is a critical consideration for web-based systems that manage user accounts and academic information. The OWASP Top 10 project identifies common web application security risks and provides industry-standard recommendations for secure software development practices. Following OWASP guidelines helps reduce vulnerabilities and improve system security [4].

Software architecture patterns also influence the maintainability and scalability of enterprise applications. Fowler emphasized the importance of structured architectural patterns for developing maintainable and extensible software systems [5]. Similarly, Sommerville highlighted the significance of software quality attributes, system reliability, maintainability, and usability in modern software engineering practices [6]. Pressman and Maxim further discussed systematic approaches to software development, requirements engineering, software design, testing, and project management that contribute to successful software projects [7].

Based on the reviewed literature and technologies, the proposed BUP Research Bridge platform adopts modern software engineering principles, secure development practices, scalable architecture, and user-centered design approaches to address the challenges associated with research collaboration, thesis opportunity discovery, and academic networking within Bangladesh University of Professionals.

11 Conclusion

BUP Research Bridge is proposed as a centralized research collaboration platform designed to connect students, faculty members, alumni, and administrators within the research ecosystem of Bangladesh University of Professionals. The platform aims to address existing challenges associated with discovering research opportunities, identifying suitable supervisors, accessing

previous research works, and establishing meaningful academic connections.

The proposed system will provide a structured environment where research-related information can be accessed efficiently through a single platform. By integrating faculty profiles, research interests, publications, thesis opportunities, alumni expertise, and previous research projects, the platform will improve transparency, accessibility, and collaboration among stakeholders.

Furthermore, the system will contribute to strengthening the research culture within the CSE and ICT departments by encouraging greater student participation in research activities and facilitating knowledge sharing between faculty members and alumni. The proposed solution provides a strong foundation for future enhancements and has the potential to become an important academic resource for the university's research community.

Therefore, the successful implementation of BUP Research Bridge is expected to support research excellence, improve collaboration, and create long-term value for students, faculty members, alumni, and the university as a whole.

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